

Clinical Trial Subject Recruitment Engine

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Requirements:

Functional Requirements:

FR01: The system should collect and validate a trial candidate's medical, and survey information required for eligibility screening. [high priority] [Type: functional]

Acceptance Criteria: **Pass:** Valid candidate information is successfully stored and missing mandatory information is reported. **Fail:** Incomplete candidate information is accepted as a valid submission.

Rationale: Accurate candidate information is necessary for reliable eligibility evaluation.

FR02: System shall validate the candidate responses against the trail's inclusion and exclusion criteria and should classify as eligible or ineligible. [high priority] [Type: functional]

Acceptance Criteria: **Pass:** An ineligible candidate is flagged with the specific failed criterion. **Fail:** A candidate violating an exclusion criterion is classified as eligible.

Rationale: Eligibility screening is the core function of the recruitment engine.

FR03: The system shall record and track informed consents for eligible candidates.

[high priority] [Type: functional]

Acceptance Criteria: **Pass:** The system records consent status and timestamp and identifies candidates whose consent is incomplete. **Fail:** A candidate without required consent is treated as fully enrolled.

Rationale: Informed consent is a mandatory step before trial participation.

FR04: The system shall record scheduled clinical-trial visits. [medium priority] [Type: functional]

Acceptance Criteria: **Pass:** Completed and missed visits are correctly displayed. **Fail:** A missed visit is incorrectly recorded as completed.

Rationale: Clinical trial visit records are required for system to process candidate to upcoming steps.

FR05: System shall allow clinical researchers to review candidate eligibility, consent status, and visit compliance records. [high priority] [Type: functional]

Acceptance Criteria: **Pass:** Researchers can retrieve the required candidate information. **Fail:** Researchers cannot access or receive incomplete recruitment records.

Rationale: Researchers need centralized information to make recruitment and monitoring decision

Non-Functional Requirements:

NF01: System should maintain the logs for documents modification, creation and access.

[high priority] [Type: security requirement]

Acceptance Criteria: **Pass:** Every event is recorded with user identity, timestamp, action, and integrity protection, and unauthorized alteration is detected. **Fail:** A record can be modified without generating an audit entry.

Rationale: Clinical-trial records are sensitive and require traceability, integrity.

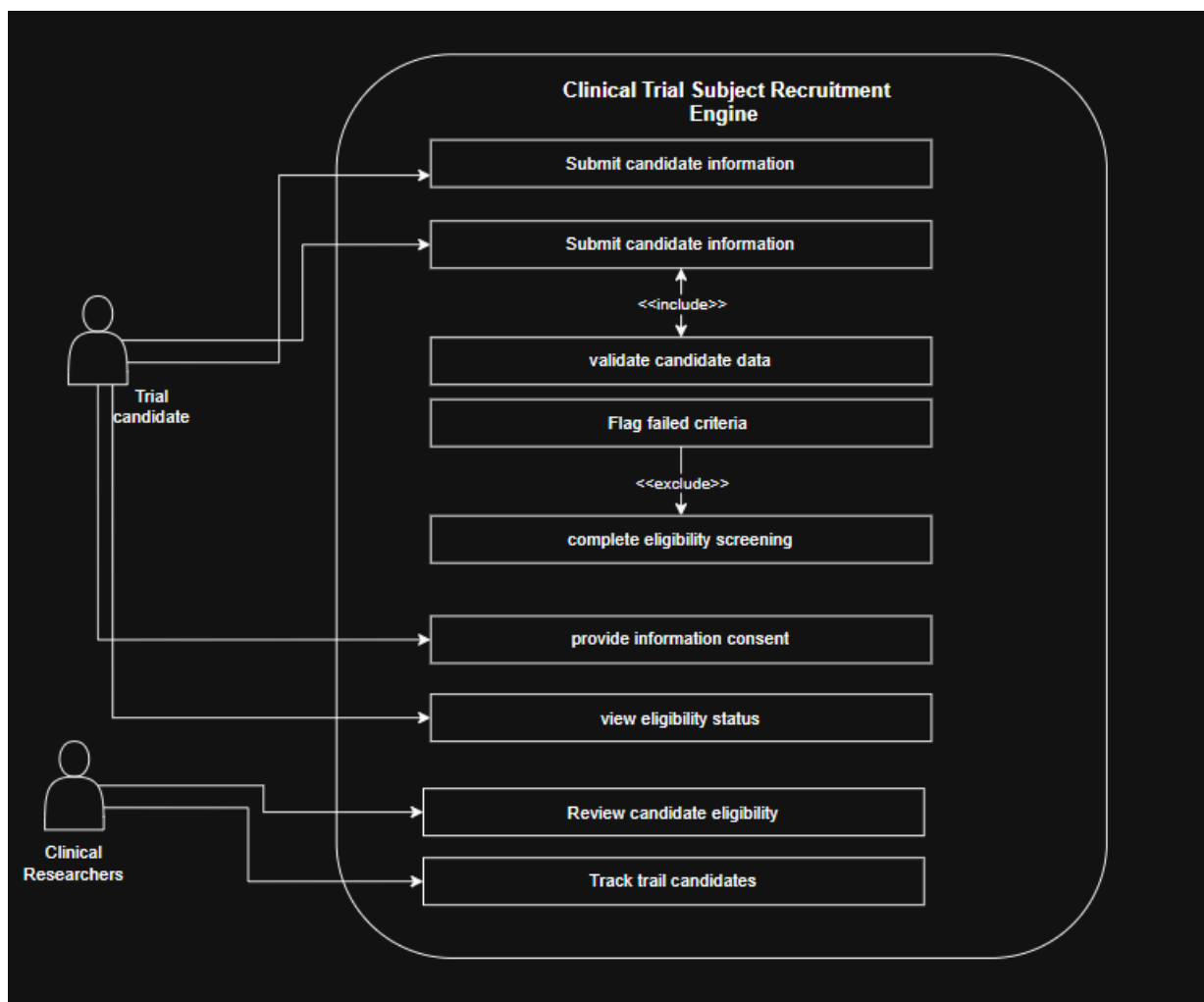
NF02: System should return candidate eligibility evaluation results within 2-5ms even under the peak load.

[high priority] [Type: performance requirement]

Acceptance Criteria: **Pass:** At least 95% of eligibility requests complete within 2 seconds during simulated peak load. **Fail:** More than 5% of requests exceed the 2-second threshold.

Rationale: Fast screening improves researcher efficiency and candidate recruitment while providing a measurable performance target.

UML:



Use Case Flow:

Use Case ID-UC-01

Use Case Name - Evaluate Candidate Eligibility

Primary Actor - Clinical Trial Candidate

Supporting Actor - Clinical Researcher

Description: The system evaluates a candidate's submitted demographic and medical information against the inclusion and exclusion criteria of a clinical trial and determines whether the candidate is eligible to participate.

Preconditions

1. A clinical trial is configured in the system.
2. Inclusion and exclusion criteria have been defined.
3. The candidate has provided the required information.
4. The candidate's submitted information passes basic validation.
- 5.

Postconditions

Success

- Candidate eligibility status is stored.
- The system records the eligibility evaluation.
- The candidate can view their eligibility status.
- The researcher can review the result.

Failure

- Invalid or incomplete information is reported.
- Eligibility is not finalized until the required information is corrected.

Main Success Scenario

1. The candidate opens the eligibility screening form.
2. The system displays the required medical questions.
3. The candidate enters the requested information.
4. The candidate submits the screening form.
5. The system validates the submitted information.
6. The system evaluates the information against the trial's inclusion criteria.
7. The system evaluates the information against the trial's exclusion criteria.
8. The system determines that the candidate satisfies all required criteria.
9. The system classifies the candidate as Eligible.
10. The system stores the eligibility result with a timestamp.
11. The candidate and clinical researcher can view the eligibility status.

Alternate Flow

Candidate Fails an Exclusion Criterion

At Step 7:

1. The system detects that the candidate satisfies an exclusion criterion.
2. The system classifies the candidate as Ineligible.
3. The system identifies the specific failed criterion.
4. The system records the eligibility decision.
5. The candidate and researcher can view the results

